

Hence,

$$\begin{aligned}
 |F'(z)| &\leq \frac{1}{2\pi} \int_{C_\delta(z)} \frac{|F(\zeta)|}{|\zeta - z|^2} |d\zeta| \\
 &\leq \frac{1}{2\pi} \sup_{\zeta \in \Omega} |F(\zeta)| \frac{1}{\delta^2} 2\pi\delta \\
 &= \frac{1}{\delta} \sup_{\zeta \in \Omega} |F(\zeta)|,
 \end{aligned}$$

as was to be shown.

Of course, there is nothing special about the first derivative, and in fact under the hypotheses of the last theorem, we may conclude (arguing as above) that for every $k \geq 0$ the sequence of k^{th} derivatives $\{f_n^{(k)}\}_{n=1}^\infty$ converges uniformly to $f^{(k)}$ on every compact subset of Ω .

In practice, one often uses Theorem 5.2 to construct holomorphic functions (say, with a prescribed property) as a series

$$(14) \quad F(z) = \sum_{n=1}^{\infty} f_n(z).$$

Indeed, if each f_n is holomorphic in a given region Ω of the complex plane, and the series converges uniformly in compact subsets of Ω , then Theorem 5.2 guarantees that F is also holomorphic in Ω . For instance, various special functions are often expressed in terms of a converging series like (14). A specific example is the Riemann zeta function discussed in Chapter 6.

We now turn to a variant of this idea, which consists of functions defined in terms of integrals.

5.3 Holomorphic functions defined in terms of integrals

As we shall see later in this book, a number of other special functions are defined in terms of integrals of the type

$$f(z) = \int_a^b F(z, s) ds,$$

or as limits of such integrals. Here, the function F is holomorphic in the first argument, and continuous in the second. The integral is taken in the sense of Riemann integration over the bounded interval $[a, b]$. The problem then is to establish that f is holomorphic.

In the next theorem, we impose a sufficient condition on F , often satisfied in practice, that easily implies that f is holomorphic.

After a simple linear change of variables, we may assume that $a = 0$ and $b = 1$.

Theorem 5.4 *Let $F(z, s)$ be defined for $(z, s) \in \Omega \times [0, 1]$ where Ω is an open set in $\mathbb{C}$. Suppose F satisfies the following properties:*

- (i) $F(z, s)$ is holomorphic in z for each s .
- (ii) F is continuous on $\Omega \times [0, 1]$.

Then the function f defined on Ω by

$$f(z) = \int_0^1 F(z, s) ds$$

is holomorphic.

The second condition says that F is jointly continuous in both arguments.

To prove this result, it suffices to prove that f is holomorphic in any disc D contained in Ω , and by Morera's theorem this could be achieved by showing that for any triangle T contained in D we have

$$\int_T \int_0^1 F(z, s) ds dz = 0.$$

Interchanging the order of integration, and using property (i) would then yield the desired result. We can, however, get around the issue of justifying the change in the order of integration by arguing differently. The idea is to interpret the integral as a “uniform” limit of Riemann sums, and then apply the results of the previous section.

Proof. For each $n \geq 1$, we consider the Riemann sum

$$f_n(z) = (1/n) \sum_{k=1}^n F(z, k/n).$$

Then f_n is holomorphic in all of Ω by property (i), and we claim that on any disc D whose closure is contained in Ω , the sequence $\{f_n\}_{n=1}^\infty$ converges uniformly to f . To see this, we recall that a continuous function on a compact set is uniformly continuous, so if $\epsilon > 0$ there exists $\delta > 0$ such that

$$\sup_{z \in D} |F(z, s_1) - F(z, s_2)| < \epsilon \quad \text{whenever } |s_1 - s_2| < \delta.$$

Then, if $n > 1/\delta$, and $z \in D$ we have

$$\begin{aligned} |f_n(z) - f(z)| &= \left| \sum_{k=1}^n \int_{(k-1)/n}^{k/n} F(z, k/n) - F(z, s) \, ds \right| \\ &\leq \sum_{k=1}^n \int_{(k-1)/n}^{k/n} |F(z, k/n) - F(z, s)| \, ds \\ &< \sum_{k=1}^n \frac{\epsilon}{n} \\ &< \epsilon. \end{aligned}$$

This proves the claim, and by Theorem 5.2 we conclude that f is holomorphic in D . As a consequence, f is holomorphic in Ω , as was to be shown.

5.4 Schwarz reflection principle

In real analysis, there are various situations where one wishes to extend a function from a given set to a larger one. Several techniques exist that provide extensions for continuous functions, and more generally for functions with varying degrees of smoothness. Of course, the difficulty of the technique increases as we impose more conditions on the extension.

The situation is very different for holomorphic functions. Not only are these functions indefinitely differentiable in their domain of definition, but they also have additional characteristically rigid properties, which make them difficult to mold. For example, there exist holomorphic functions in a disc which are continuous on the closure of the disc, but which cannot be continued (analytically) into any region larger than the disc. (This phenomenon is discussed in Problem 1.) Another fact we have seen above is that holomorphic functions must be identically zero if they vanish on small open sets (or even, for example, a non-zero line segment).

It turns out that the theory developed in this chapter provides a simple extension phenomenon that is very useful in applications: the Schwarz reflection principle. The proof consists of two parts. First we define the extension, and then check that the resulting function is still holomorphic. We begin with this second point.

Let Ω be an open subset of $\mathbb{C}$ that is symmetric with respect to the real line, that is

$$z \in \Omega \quad \text{if and only if} \quad \bar{z} \in \Omega.$$